

Monte-Carlo simulation

Sampling and estimation

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When does sampling occur?

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- The i.i.d. random stiffnesses of a beam (lecture #1);
- Classical r.v.: exponential, Gamma, β , β' , log-normal... (lecture #2);
- Polynomial chaos expansion for the representation of second-order r.v. (lecture #2);
- $\mathbf{X} \sim p_{\mathbf{X}}(\mathbf{x}) = e^{-\lambda_0 - \lambda \cdot \mathbf{C}(\mathbf{x})}$ derived from the MaxEnt principle (lecture #2);
- Karhunen-Loève expansion of second-order random processes (lecture #3);
- Spectral expansion of stationary second-order random processes (lecture #4).

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Definition

Uniform probability law $\mathcal{U}(0, 1) = \mathbf{1}_{[0,1]}(u)du$ and:

$$m_U = \underline{U} = \frac{1}{2}, \quad \sigma_U = \sqrt{\mathbb{E}\{(U - m_U)^2\}} = \frac{1}{2\sqrt{3}}.$$

Theorem

Let F be the distribution function of a continuous r.v. X . Then if $U \sim \mathcal{U}(0, 1)$ the r.v. $F^{-1}(U)$ has distribution function F and the r.v. $F(X)$ has a uniform probability law.

- $P(F^{-1}(U) \leq x) = P(U \leq F(x)) = \int_0^{F(x)} du = F(x).$
- $P(F(X) \leq u) = P(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u$ with the generalized inverse (or quantile) function:

$$F^{-1}(u) = \inf\{x | F(x) \geq u\}, \quad 0 < u < 1.$$

Inverse transform method

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PDF	F	$X = F^{-1}(U)$	Equivalent form
Exponential ($\sigma > 0$) $\frac{1}{\sigma} e^{-\frac{x}{\sigma}} \mathbf{1}_{\mathbb{R}_+}(x)$	$1 - e^{-\frac{x}{\sigma}}$	$-\sigma \ln(1 - U)$	$-\sigma \ln(U)$
Weibull ($\sigma, k > 0$) $\frac{k}{\sigma} (\frac{x}{\sigma})^{k-1} e^{-(\frac{x}{\sigma})^k} \mathbf{1}_{\mathbb{R}_+}(x)$	$1 - e^{-(\frac{x}{\sigma})^k}$	$\sigma(-\ln(1 - U))^{\frac{1}{k}}$	$\sigma(-\ln(U))^{\frac{1}{k}}$
Cauchy $\frac{\sigma}{\pi[\sigma^2 + (x - \mu)^2]}$	$\frac{1}{2} + \frac{1}{\pi} \arctan(\frac{x - \mu}{\sigma})$	$\mu + \sigma \tan \pi(U - \frac{1}{2})$	$\mu + \sigma \tan(\pi U)$
Rayleigh $\frac{x}{\sigma} e^{-\frac{x^2}{2\sigma^2}} \mathbf{1}_{\mathbb{R}_+}(x)$	$1 - e^{-\frac{x^2}{2\sigma^2}}$	$\sigma \sqrt{-\ln(1 - U)}$	$\sigma \sqrt{-\ln(U)}$
Gamma ($k \in \mathbb{N}^*$) $\frac{1}{\sigma \Gamma(k)} (\frac{x}{\sigma})^{k-1} e^{-\frac{x}{\sigma}} \mathbf{1}_{\mathbb{R}_+}(x)$	$\int_0^{\frac{x}{\sigma}} \frac{t^{k-1}}{\Gamma(k)} e^{-t} dt$	$-\sigma \sum_{j=1}^k \ln U_j$	
Triangular $\frac{2}{\sigma} (1 - \frac{x}{\sigma}) \mathbf{1}_{[0, \sigma]}(x)$	$\frac{2}{\sigma} (x - \frac{x^2}{2\sigma}) \mathbf{1}_{[0, \sigma]}(x)$	$\sigma(1 - \sqrt{1 - U})$	$\sigma(1 - \sqrt{U})$

The case of Gaussian r.v.

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- Let $G \sim \mathcal{N}(0, 1)$, the distribution function is the error function (erf), which is difficult to inverse:

$$F(g) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^g e^{-\frac{x^2}{2}} dx.$$

- *Box-Muller algorithm*: Let $U_1, U_2 \sim \mathcal{U}(0, 1)$ independent and

$$G_1 = \sqrt{-2 \ln U_1} \cos(2\pi U_2),$$

$$G_2 = \sqrt{-2 \ln U_1} \sin(2\pi U_2),$$

then G_1 and G_2 are i.i.d. normal r.v. $G_1, G_2 \sim \mathcal{N}(0, 1)$.

By causality principle $p_G(g) = p_U(h^{-1}(g)) \det(\nabla_g h^{-1}(g))$, and the inverse of $u \mapsto h(u)$ is $h^{-1}(g) = (\exp(-\|g\|^2/2), \frac{1}{2\pi} \arctan(g_2/g_1))$.

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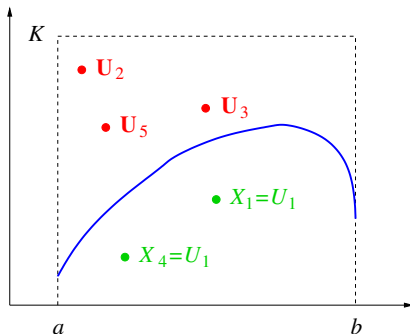
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- Objective: simulate $X \sim \pi(x)dx$ where $\pi(x)$ has compact support within $[a, b]$.
- Basic algorithm for $0 < \sup_{x \in [a, b]} \pi(x) \leq K$:
 - 1 $U \sim \mathcal{U}([a, b] \times [0, K])$ a uniform r.v. on $[a, b] \times [0, K]$;
 - 2 if $U_2 \leq \pi(U_1)$ then $X = U_1$, else goto 1.



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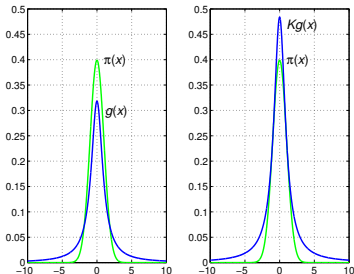
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- Generalization for $0 < \pi(x) \leq Kg(x)$ where g is a known PDF s.t. $g \approx \pi$ and $G \sim g(x)dx$ is easy to simulate:
 - 1 $U \sim \mathcal{U}(0, 1)$ and $G \sim g(x)$ independently;
 - 2 if $K \times U \times g(G) < \pi(G)$ then $X = G$, else goto 1.
- The optimum choice $K = \sup_x \frac{\pi(x)}{g(x)}$, and K^{-1} is the probability of acceptance.

Example: simulate a normal r.v. from a Cauchy r.v.



Target: $\pi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$
Proposal: $g(x) = [\pi(1+x^2)]^{-1}$,
then $K = \sqrt{2\pi}/e$ and use
 $G = \tan(\pi U)$ with $U \sim \mathcal{U}(0, 1)$.

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Generate a sequence of pseudo-random numbers of which distribution is close to $\mathcal{U}(0, 1)$ with:

- good statistical properties and uniformness;
- long periods;
- efficiency vs. computational time;
- repeatability (in order to test programs);
- ease of implementation for all programming languages;
- unpredictability: it should be impossible to infer U_i from the knowledge of U_{i-1} (applications in cryptography).

Pseudo-random number generators PRNGs

Linear Congruential Generator

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- Pseudo-random numbers $\{U_i\}_{0 \leq i \leq m-1}$ following a uniform law may be obtained by a *Linear Congruential Generator* $\text{LCG}(a, b, m)$:

$$I_i = (aI_{i-1} + b) \bmod(m), \quad I_0 = \text{SEED},$$
$$U_i = \frac{I_i}{m},$$

with $0 < a < m$ (multiplier), $0 \leq b < m$ (increment),
 $0 < m$ (modulus), $0 \leq I_0 < m$ (SEED).

Example: $a = 7^5$, $b = 0$ (*Multiplicative* Congruential Generator),
 $m = 2^{31} - 1$ is frequently used.

- For $\mathbf{U} \in [0, 1]^n$ with mutually independent components use n independent versions of U .

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Multiple Recursive Generator

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Estimation

- Pseudo-random numbers $\{U_i\}_{0 \leq i \leq m-1}$ following a uniform law may be obtained by *Multiple Recursive Generator* MRG(k, m):

$$I_i = (a_1 I_{i-1} + \cdots + a_k I_{i-k}) \bmod(m),$$
$$U_i = \frac{I_i}{m},$$

where $a_k \in \{-(m-1), \dots, (m+1)\}$.

- **Properties:**
 - The period is greater than m and the maximum period is $m^k - 1$ whenever m is a prime number;
 - Generalization to the multi-dimensional case.

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Other examples

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- *Combined Multiple Recursive Generator MRG32k3a*
(L'Écuyer 1999):

$$I_i = (1403580I_{i-2} - 810728I_{i-3}) \bmod(2^{32} - 209),$$

$$J_i = (527612J_{i-1} - 1370589J_{i-3}) \bmod(2^{32} - 22853),$$

$$U_i = \frac{(I_i - J_i) \bmod(2^{32} - 209)}{2^{32} - 209}.$$

- *Mersenne Twister algorithm* (because its period is the Mersenne number $2^{19937} - 1$, see Matsumoto-Nishimura 1997):
<http://www.math.sci.hiroshima-u.ac.jp/~m-mat/MT/emt.html>

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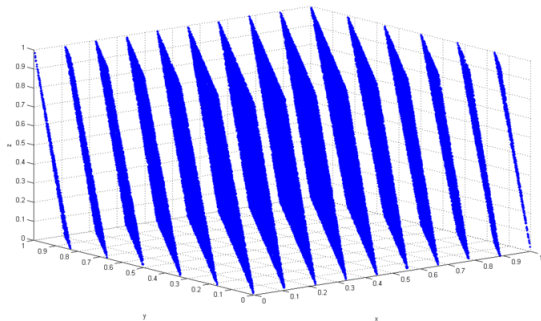
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■ RANDU (IBM 1967):

$$I_i = 65539I_{i-1} \bmod(2^{31}),$$

$$U_i = \frac{I_i}{2^{31}},$$

with I_0 odd.



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Markov chains

Basic definition

Definition

- A Markov¹ chain $(\mathbf{U}_m, m \in \mathbb{N})$ is a sequence of r.v. with values in a finite or countable set E defined by:

- $E = \{\mathbf{u}_0, \mathbf{u}_1, \dots\},$

$$\begin{aligned} P(\mathbf{U}_{m+1} = \mathbf{u}_{m+1} | \mathbf{U}_0 = \mathbf{u}_0, \mathbf{U}_1 = \mathbf{u}_1, \dots, \mathbf{U}_m = \mathbf{u}_m) \\ = P(\mathbf{U}_{m+1} = \mathbf{u}_{m+1} | \mathbf{U}_m = \mathbf{u}_m), \quad \forall m \in \mathbb{N}; \end{aligned}$$

- its initial probability law $\pi_{0,j} = P(\mathbf{U}_0 = \mathbf{u}_j).$
- A homogeneous Markov chain:

$$\begin{aligned} P(\mathbf{U}_{m+1} = \mathbf{u}_j | \mathbf{U}_m = \mathbf{u}_i) \\ = P(\mathbf{U}_m = \mathbf{u}_j | \mathbf{U}_{m-1} = \mathbf{u}_i), \quad \forall m \in \mathbb{N}. \end{aligned}$$

¹ Andreï Markov (1856–1922): Russian mathematician.

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- Let $(V_m, m \in \mathbb{N}^*)$ be a sequence of i.i.d. r.v. with values in F , and let $f : E \times F \rightarrow E$ be measurable. Then the sequence $(U_m, m \in \mathbb{N})$ defined by:

$$U_{m+1} = f(U_m, V_{m+1}), \quad \forall m \in \mathbb{N},$$

is an homogeneous Markov chain as soon as V_m is independent of U_0 .

- Random walk in $E = \mathbb{Z}^2$:

$$U_{m+1} = U_m + V_{m+1}, \quad \forall m \in \mathbb{N},$$

with $V_m \in F = \{(-1, 0), (0, -1), (+1, 0), (0, +1)\}$.

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Estimation

■ Sampling a Markov chain:

1 $U_0 \sim \pi_0$;

2 repeat

■ $U_{m-1} = \mathbf{u}_i$

■ $U \sim \mathcal{U}(0, 1)$ and find j s.t.:

$$\sum_{k=1}^{j-1} P(U_m = \mathbf{u}_k | U_{m-1} = \mathbf{u}_i) \leq U < \sum_{k=1}^j P(U_m = \mathbf{u}_k | U_{m-1} = \mathbf{u}_i)$$

■ $U_m = \mathbf{u}_j$

until $m = m_{\text{final}}$.

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Transition kernel and invariant measure

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- A Markov chain is fully characterized by its initial distribution π_0 and its *transition kernel* $\Pi(m)$ s.t.:

$$\Pi(m) = [\pi_{ij}(m)]_{i,j \in E}, \quad \pi_{ij}(m) = P(\mathbf{U}_m = \mathbf{u}_j | \mathbf{U}_{m-1} = \mathbf{u}_i).$$

Remarks:

- Π is independent of m if the Markov chain is homogeneous.
- π_{ii} is not necessarily zero.
- $\sum_{j \in E} \pi_{ij}(m) = 1, \forall i \in E.$
- An *invariant measure* (or *stationary distribution*)

$\pi^* = (\pi_j^*)_{j \in E}$ is s.t.:

$$\pi_j^* \geq 0, \quad \pi_j^* = \sum_{i \in E} \pi_i^* \pi_{ij}(m), \quad \forall j \in E, \forall m \in \mathbb{N},$$

with the normalization $\sum_{j \in E} \pi_j^* = 1.$

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Classification of the states

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Definition

- *A Markov chain is irreducible if it is possible to reach any state from any other state.*
- *The state $j \in E$ is transient if, starting from it, the probability it is never reached again is non-zero; otherwise it is recurrent.*
- *A recurrent state is positive recurrent if the mean return time is finite: $\mathbb{E}\{\inf\{k \geq 1 | U_k = \mathbf{u}_j\} | U_0 = \mathbf{u}_j\} < +\infty$.*
- *The state $j \in E$ is aperiodic if, starting from it, it can be reached again at any subsequent time with a non-zero probability.*

If $\pi_{ij} > 0 \forall i, j \in E$, then the MC is irreducible and aperiodic.

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Existence of an invariant measure for homogeneous MC

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Theorem

- *If a MC has at least one positive recurrent state, it has an invariant measure. It is unique if in addition the MC is irreducible.*
- *If a MC is irreducible, positive recurrent and aperiodic, then there exists a unique invariant measure π^* such that $(U_m) \xrightarrow[m \rightarrow +\infty]{\mathcal{L}} \pi^*$ independently of π_0 .*
- The latter result is the convergence of the k -stage transition Π^k through the kernel Π to the invariant measure:
$$\Pi^k = [\pi_{ij}^{(k)}]_{i,j \in E}, \quad \pi_{ij}^{(k)} = P(U_{m+k} = \mathbf{u}_j | U_m = \mathbf{u}_i),$$
such that if $\pi_k = P(U_k = \mathbf{u}_j)_{j \in E}$ then $\pi_k = \pi_0 \Pi^k$.
- If π_0 is an invariant measure, then $\pi_k = \pi_0 \forall k \in \mathbb{N}$.

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Theorem

Let $(\mathbf{U}_m, m \in \mathbb{N})$ be an irreducible, positive recurrent homogeneous MC. Then there exists an invariant measure π^ such that for all regular functions f :*

$$\lim_{m \rightarrow +\infty} \mathbb{E}\{f(\mathbf{U}_m)\} = \lim_{m \rightarrow +\infty} \frac{1}{m} \sum_{k=0}^{m-1} f(\mathbf{U}_k) = \sum_{j \in E} \pi_j^* f(\mathbf{u}_j) \text{ a.s.}$$

- The ergodic theorem is an extension of the *Law of Large Numbers* to homogeneous MCs: let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_m$ be m i.i.d. r.v. with the same law $P_{\mathbf{X}}$, then for all regular functions f :

$$\lim_{m \rightarrow +\infty} \frac{1}{m} \sum_{k=1}^m f(\mathbf{X}_k) = \mathbb{E}\{f(\mathbf{X})\} = \int f(\mathbf{x}) P_{\mathbf{X}}(\mathrm{d}\mathbf{x}) \text{ a.s.}$$

- It allows to estimate the invariant measure from one sample chain.

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Proposition

Let $\pi(\mathbf{u}_i, \mathbf{u}_j) = p(\mathbf{u}_i, \mathbf{u}_j) + r(\mathbf{u}_i)\delta(\mathbf{u}_j - \mathbf{u}_i)$ where:

- $p(\mathbf{u}_i, \mathbf{u}_i) = 0$ and a reversibility condition holds:

$$\pi^*(\mathbf{u}_i)p(\mathbf{u}_i, \mathbf{u}_j) = \pi^*(\mathbf{u}_j)p(\mathbf{u}_j, \mathbf{u}_i), \quad \forall i, j \in E;$$

- $r(\mathbf{u}_i) = 1 - \sum_{j \in E} p(\mathbf{u}_i, \mathbf{u}_j)$.

Then $\pi^*(\mathbf{u})$ is the invariant density of the MC of which transition kernel is $\Pi = [\pi(\mathbf{u}_i, \mathbf{u}_j)]_{i,j \in E}$.

$$\begin{aligned} \sum_{i \in E} \pi^*(\mathbf{u}_i)\pi(\mathbf{u}_i, \mathbf{u}_j) &= \sum_{i \in E} \pi^*(\mathbf{u}_i)p(\mathbf{u}_i, \mathbf{u}_j) + \sum_{i \in E} \pi^*(\mathbf{u}_i)r(\mathbf{u}_i)\delta(\mathbf{u}_j - \mathbf{u}_i) \\ &= \sum_{i \in E} \pi^*(\mathbf{u}_j)p(\mathbf{u}_j, \mathbf{u}_i) + \pi^*(\mathbf{u}_j)r(\mathbf{u}_j) \\ &= \pi^*(\mathbf{u}_j)(1 - r(\mathbf{u}_j)) + \pi^*(\mathbf{u}_j)r(\mathbf{u}_j) \\ &= \pi^*(\mathbf{u}_j). \end{aligned}$$

MCMC methods

Metropolis-Hastings algorithm (1953, 1970)

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Estimation

- Objective: simulate $\mathbf{X} \sim C^* \pi^*(\mathbf{x}) d\mathbf{x}$, $E \equiv \Omega \subseteq \mathbb{R}^n$.
- Let $q(\mathbf{x}, \mathbf{y}) \geq 0$ be a *candidate* (or *instrumental* or *proposal*) density s.t. $\text{supp } \pi^*(\cdot) \subset \text{supp } q(\mathbf{x}, \cdot)$ and define the *probability of move*:

$$\alpha(\mathbf{x}, \mathbf{y}) = \min \left\{ \frac{\pi^*(\mathbf{y})q(\mathbf{y}, \mathbf{x})}{\pi^*(\mathbf{x})q(\mathbf{x}, \mathbf{y})}, 1 \right\}.$$

1 $\mathbf{x}_0 \sim \pi_0(\mathbf{x}) d\mathbf{x}$

2 repeat

■ $\mathbf{X}_m = \mathbf{x}_m$

■ $\mathbf{Y} \sim q(\mathbf{x}_m, \mathbf{y}) d\mathbf{y}$ and $U \sim \mathcal{U}(0, 1)$

■ if $U \leq \alpha(\mathbf{x}_m, \mathbf{Y})$ then $\mathbf{X}_{m+1} = \mathbf{Y}$, else $\mathbf{X}_{m+1} = \mathbf{X}_m$

until $m = m_{\text{final}}$.

Assume $\pi^*(\mathbf{x})q(\mathbf{x}, \mathbf{y}) > \pi^*(\mathbf{y})q(\mathbf{y}, \mathbf{x})$, then $\alpha(\mathbf{x}, \mathbf{y})$ is tuned so that $\alpha(\mathbf{x}, \mathbf{y})q(\mathbf{x}, \mathbf{y})$ satisfies the reversibility condition.

MCMC methods

Application to the MaxEnt

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Estimation

- The MaxEnt principle leads to the non-linear optimization problem of finding $\boldsymbol{\lambda} \in \mathbb{R}^\ell$ s.t.:

$$\boldsymbol{g}(\boldsymbol{\lambda}) := \int_{\mathbb{R}^n} (\boldsymbol{C}(\boldsymbol{x}) - \underline{\boldsymbol{C}}) e^{-\boldsymbol{\lambda} \cdot (\boldsymbol{C}(\boldsymbol{x}) - \underline{\boldsymbol{C}})} d\boldsymbol{x} = \mathbf{0}.$$

- By the ergodic theorem:

$$\boldsymbol{g}(\boldsymbol{\lambda}) \simeq \lim_{m \rightarrow +\infty} \frac{1}{m} \sum_{k=0}^{m-1} (\boldsymbol{C}(\boldsymbol{X}_k(\theta)) - \underline{\boldsymbol{C}}),$$

and by the law of large number for $M \in \mathbb{N}$ large enough:

$$\boldsymbol{g}(\boldsymbol{\lambda}) \simeq \lim_{n_s \rightarrow +\infty} \frac{1}{n_s} \sum_{k=1}^{n_s} (\boldsymbol{C}(\boldsymbol{X}_M(\theta_k)) - \underline{\boldsymbol{C}}),$$

where $(\boldsymbol{X}_m, m \in \mathbb{N})$ is the MC of which invariant density is $\pi^*(\boldsymbol{x}) = e^{-\boldsymbol{\lambda} \cdot (\boldsymbol{C}(\boldsymbol{x}) - \underline{\boldsymbol{C}})}$.

When does estimation occur?

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Estimation

- Parametrization of *e.g.* Gamma or β distributions (Lamé's moduli, Poisson's coefficient):

- If $X \sim \Gamma(k, \frac{1}{\sigma})$ then

$$\underline{X} = k\sigma, \sigma_X^2 = k\sigma^2, s_X = \frac{2}{\sqrt{k}}, \kappa_X = \frac{6}{k}, \text{ etc.}$$

- If $X \sim \beta(k, \frac{1}{\sigma})$ then

$$\underline{X} = \frac{k\sigma}{1+k\sigma}, \sigma_X^2 = \frac{k\sigma^2}{(1+k\sigma)^2(1+\sigma+k\sigma)}, \text{ etc.}$$

- $p_X(\mathbf{x}) = e^{-\lambda_0 - \boldsymbol{\lambda} \cdot \mathbf{C}(\mathbf{x})}$ where $(\lambda_0, \boldsymbol{\lambda})$ depend on $\underline{\mathbf{C}}$.

Estimator

Punctual & sequential estimation

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Estimation

- Let $\mathbf{X}(\theta)$ be a second order r.v. defined on $(\Omega_\theta, \mathcal{E}, P)$ with values in $\Omega_{\mathbf{X}} \subseteq \mathbb{R}^n$ and probability distribution $P_{\mathbf{X}}(d\mathbf{x}; \boldsymbol{\nu})$ depending on (deterministic) parameters $\boldsymbol{\nu} \in \Theta \subseteq \mathbb{R}^p$.
 - Sequential or continuous estimation: estimate $\boldsymbol{\nu}$ from M independent realizations $\mathbf{x}^{(m)} = \mathbf{X}(\theta_m)$, $\theta_m \in \Omega_\theta$, $1 \leq m \leq M$;
 - Punctual estimation: $M = 1$.
- An estimation is a (measurable) rule $\mathbf{x}_M \mapsto \hat{\boldsymbol{\nu}}_M(\mathbf{x}_M)$, $\mathbf{x}_M = (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(M)})$, for calculating the *estimate* $\hat{\boldsymbol{\nu}}_M(\mathbf{x}_M)$. The corresponding *estimator* is the r.v. $\hat{\boldsymbol{\nu}}_M(\mathbf{X}_M)$.
- The *estimation error* is $\boldsymbol{\epsilon}_M(\mathbf{X}_M) = \hat{\boldsymbol{\nu}}_M(\mathbf{X}_M) - \boldsymbol{\nu}$.
- Two classes of methods for building estimators:
 - (i) method of moments and (ii) maximum likelihood.

Estimator

Characterizing an estimator

Definition

- The bias of the estimator $\mathbf{b}_M(\boldsymbol{\nu}) = \mathbb{E}\{\boldsymbol{\epsilon}_M(\mathbf{X}_M)\}$. It is unbiased if $\mathbf{b}_M = \mathbf{0} \forall M$, or asymptotically unbiased if $\lim_{M \rightarrow +\infty} \mathbf{b}_M = \mathbf{0}$.
- The dispersion of the estimator:

$$\begin{aligned} \mathbf{V}_M(\boldsymbol{\nu}) &= \mathbb{E}\{\boldsymbol{\epsilon}_M(\mathbf{X}_M) \otimes \boldsymbol{\epsilon}_M(\mathbf{X}_M)\} \\ &= \mathbf{C}_{\hat{\boldsymbol{\nu}}_M} + \mathbf{b}_M(\boldsymbol{\nu}) \otimes \mathbf{b}_M(\boldsymbol{\nu}). \end{aligned}$$

The lower $\text{Tr } \mathbf{C}_{\hat{\boldsymbol{\nu}}_M}$ is, the more efficient it is.

- The symmetric, positive Fisher information matrix:

$$[\mathcal{I}_M(\boldsymbol{\nu})] = \mathbb{E}\{\boldsymbol{\nabla}_{\boldsymbol{\nu}} \ln \mathcal{L}(\boldsymbol{\nu}|\mathbf{X}_M) \otimes \boldsymbol{\nabla}_{\boldsymbol{\nu}} \ln \mathcal{L}(\boldsymbol{\nu}|\mathbf{X}_M)\},$$

for $\mathcal{L}(\boldsymbol{\nu}|\mathbf{x}_M) := \prod_{m=1}^M p_{\mathbf{X}}(\mathbf{x}^{(m)}; \boldsymbol{\nu})$, $p_{\mathbf{X}} = \frac{dP_{\mathbf{X}}}{d\mathbf{x}}$.

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Cramér-Rao inequality and efficiency

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The Cramér-Rao inequality for the j -th parameter:

$$[\mathbf{V}_M(\boldsymbol{\nu})]_{jj} = \mathbb{E}\{\epsilon_{M,j}^2(\mathbf{X}_M)\} \geq \frac{(1 + \partial_{\nu_j} b_{M,j}(\boldsymbol{\nu}))^2}{[\mathcal{I}_M(\boldsymbol{\nu})]_{jj}},$$

with an equality iff $\partial_{\nu_j} \ln \mathcal{L}(\boldsymbol{\nu}|\mathbf{x}_M) = \varphi_j(\boldsymbol{\nu})\epsilon_{M,j}(\mathbf{x}_M)$.

- An estimator is *efficient* if it minimizes its dispersion, thus iff the above condition holds, or:

$$\mathbb{E}\{\partial_{\nu_j} \ln \mathcal{L}(\boldsymbol{\nu}|\mathbf{X}_M)\} = \varphi_j(\boldsymbol{\nu})b_{M,j}(\boldsymbol{\nu}).$$

- If it is unbiased $[\mathbf{V}_M(\boldsymbol{\nu})]_{jj} \geq [\mathcal{I}_M(\boldsymbol{\nu})]_{jj}^{-1}$ (Cramér-Rao lower bound) and

$$\mathbb{E}\{\partial_{\nu_j} \ln \mathcal{L}(\boldsymbol{\nu}|\mathbf{X}_M)\} = 0.$$

Maximum likelihood method

ML estimator

Definition

- Consider the following estimate $\hat{\ell}(\boldsymbol{\nu})$ of the expected log-likelihood $\ell(\boldsymbol{\nu}) = \mathbb{E}\{\ln p_{\mathbf{X}}(\mathbf{X}; \boldsymbol{\nu})\}$ of a single observation:

$$\hat{\ell}(\boldsymbol{\nu}) = \frac{1}{M} \ln \mathcal{L}(\boldsymbol{\nu} | \mathbf{x}_M).$$

- The maximum likelihood (ML) estimator of $\boldsymbol{\nu}$ is:

$$\hat{\boldsymbol{\nu}}_{\text{ML}}(\mathbf{x}_M) = \arg \max_{\boldsymbol{\nu} \in \Theta} \hat{\ell}(\boldsymbol{\nu}).$$

The likelihood equations relative to the realizations $\mathbf{x}_M$:

$$\partial_{\nu_j} \ln \mathcal{L}(\boldsymbol{\nu} | \mathbf{x}_M) = \sum_{m=1}^M \partial_{\nu_j} \ln p_{\mathbf{X}}(\mathbf{x}^{(m)}; \boldsymbol{\nu}) = 0, \quad 1 \leq j \leq p.$$

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Estimation

- Consistency: assume $p_{\mathbf{X}}(\cdot; \boldsymbol{\nu}_1) \neq p_{\mathbf{X}}(\cdot; \boldsymbol{\nu}_2)$ iff $\boldsymbol{\nu}_1 \neq \boldsymbol{\nu}_2$, the MLE is asymptotically unbiased and:

$$\hat{\boldsymbol{\nu}}_{\text{ML}}(\mathbf{X}_M) \xrightarrow[M \rightarrow +\infty]{\mathcal{P}} \boldsymbol{\nu}.$$

- Asymptotic normality:

$$\sqrt{M} (\hat{\boldsymbol{\nu}}_{\text{ML}}(\mathbf{X}_M) - \boldsymbol{\nu}) \xrightarrow[M \rightarrow +\infty]{\mathcal{L}} \mathcal{N}(0, [\mathcal{I}_1(\boldsymbol{\nu})]^{-1}).$$

- Efficiency: it achieves the Cramér-Rao lower bound asymptotically. Conversely, if an efficient unbiased estimator exists, then it is the MLE and it is unique.
- Functional invariance: $\varphi(\hat{\boldsymbol{\nu}}_{\text{ML}})$ is the MLE of $\varphi(\boldsymbol{\nu})$.

Maximum likelihood method

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Estimation

Consider $X \sim \mathcal{N}(\mu, \sigma^2)$ in $\mathbb{R}$, i.e. $p_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$, with $\nu_1 = \mu$ and $\nu_2 = \sigma^2$.

- The likelihood function is:

$$\mathcal{L}(\boldsymbol{\nu} | x_M) = \left(\frac{1}{2\pi\sigma^2} \right)^{\frac{M}{2}} \exp \left(-\frac{1}{2\sigma^2} \sum_{m=1}^M \left(x^{(m)} - \mu \right)^2 \right).$$

- The ML estimator is:

$$\hat{\mu}_{\text{ML}}(X_M) = \frac{1}{M} \sum_{m=1}^M X^{(m)},$$

$$\hat{\sigma}_{\text{ML}}^2(X_M) = \frac{1}{M} \sum_{m=1}^M \left(X^{(m)} - \frac{1}{M} \sum_{n=1}^M X^{(n)} \right)^2.$$

- Then $b_{M,1} = 0$, $b_{M,2} = -\frac{\sigma^2}{M}$, but is asymptotically unbiased.

Maximum likelihood method

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Estimation

Consider $X \sim \Gamma(k, \frac{1}{\sigma})$ on $\mathbb{R}_+$ with $\nu = \sigma$:

$$p_X(x) = \frac{1}{\sigma \Gamma(k)} \left(\frac{x}{\sigma}\right)^{k-1} e^{-\frac{x}{\sigma}} \mathbf{1}_{\mathbb{R}_+}(x),$$

then $\underline{X} = k\sigma = k\nu$.

- The log-likelihood function is:

$$\frac{\ln \mathcal{L}(\nu | x_M)}{M} = \frac{k-1}{M} \sum_{m=1}^M \ln x^{(m)} - \frac{1}{\sigma M} \sum_{m=1}^M x^{(m)} - k \ln \sigma - \ln \Gamma(k).$$

- The (unbiased) ML estimator is:

$$k\hat{\sigma}_{\text{ML}}(X_M) = \frac{1}{M} \sum_{m=1}^M X^{(m)}.$$

Maximum likelihood method

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Estimation

Consider the same law but now $\nu_1 = \sigma$ and $\nu_2 = k$, thus $\underline{X} = k\sigma = \nu_1\nu_2$ and $\sigma_X^2 = k\sigma^2 = \nu_1^2\nu_2$.

- The log-likelihood function is unchanged.
- But $\hat{\sigma}_{\text{ML}}$ and $\hat{k}_{\text{ML}}$ are given by a system of *non-linear* equations:

$$\hat{k}_{\text{ML}}(X_M)\hat{\sigma}_{\text{ML}}(X_M) = \frac{1}{M} \sum_{m=1}^M X^{(m)},$$

$$\psi_0(\hat{k}_{\text{ML}}(X_M)) = \frac{1}{M} \sum_{m=1}^M \ln \left(\frac{X^{(m)}}{\hat{\sigma}_{\text{ML}}(X_M)} \right),$$

where $z \mapsto \psi_0(z) = (\ln \Gamma(z))'$ is the diGamma function.

Exhaustive estimator

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Estimation

- The probability law of the estimator $\hat{\nu}_M(\mathbf{X}_M)$ should depend on ν if it is used to retrieve some information on the parameters.
- It is called *exhaustive* if it conserves the information on ν contained in the sample $\mathbf{x}_M$, *i.e.*
 $P(\mathbf{X}_M | \hat{\nu}_M(\mathbf{X}_M) = \hat{\nu})$ does not depend on ν .

Definition

$\hat{\nu}_M(\mathbf{X}_M)$ is *exhaustive* iff $\mathcal{L}(\nu | \mathbf{x}_M) = \rho_t(\mathbf{x}_M | \hat{\nu})g(\hat{\nu}; \nu)$, where ρ_t is the conditional PDF of $\mathbf{X}_M$ provided that $\hat{\nu}_M(\mathbf{X}_M) = \hat{\nu}$, and g is its PDF.

Exhaustive estimator

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Theorem

A sample $\mathbf{X}_M$ s.t. $\text{supp } p_{\mathbf{X}}$ does not depend on $\boldsymbol{\nu}$ admits an exhaustive estimator iff:

$$p_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\nu}) = e^{a(\mathbf{x})\alpha(\boldsymbol{\nu}) + b(\mathbf{x}) + \beta(\boldsymbol{\nu})} .$$

Further reading...

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